

Multi-Agent System: Theories and Applications

Lecture 1: Introduction

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Outline

- 1 Course Introduction
- 2 Multi-Agent System Examples
- 3 Terminologies and Definitions

Protection Rules for COVID-19

- **Always wear mask.**
- **Hand disinfection.**
- **Ventilation.**
- **Don't go to class if you have fever, cough, etc.**
- **Social distancing.**

Schedule

COURSE SYLLABUS

MMA Special Lectures I

Multi-Agent System: Theories and Applications

- **Prerequisite:** Linear Algebra, Linear Dynamical Systems, Programming and Simulation
 - **Final exam:** Report
 - **Period:** April 9 – July 9, 2021
 - **Lecturer:** Asst. Prof. Nguyen Dinh Hoa.
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Week	Date & Time	Lecture content	Note
1	4月9日	Course introduction, terminologies and definitions, examples in nature and engineering	
2	4月16日	Matrix theory	
3	4月23日	Stability theory of dynamical systems	
4	4月30日	Graph theory	
5	5月7日	Graph description and mathematical models of multi-agent systems	
6	5月14日	- Brief descriptions of main problems: consensus, synchronization, formation, alignment, and rendezvous - Applications in power grid, biology, robotics, transportation, aerospace, and mechanical engineering	
7	5月21日	Consensus control for first and second order linear multi-agent systems, and synchronization of linearly coupled linear oscillators	
8	5月28日	Edge dynamics	
9	6月4日	Formation control for first and second order linear multi-agent systems	
10	6月11日	Synchronization of nonlinearly coupled oscillators: Kuramoto model and its generalization	Start distributing list of topics for final report
11	6月18日	Biological oscillations: synchronization of linearly coupled Goodwin-type oscillators	
12	6月25日	Lotka-Volterra population models	Start of 1-month period for doing final report
13	7月2日	Swarm models: Vicsek, Reynold, Cucker-Smale	
14	7月9日	- Insights on analogous pairs of physical variables in different practical multi-agent systems - A glimpse of more advanced topics	
15	7月16日	Reserved week	
	7月23日	Deadline for submitting final report	

Evaluation Method: Final report

A set of problems will be given at the 12th week, then each student chooses one problem, solve it, write a report, and submit to me through email during the 16th week.

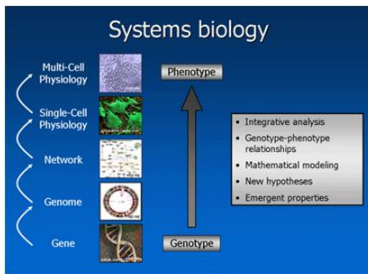
(Same problem for different students is prohibited!)

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Examples in Nature



Examples in Engineering



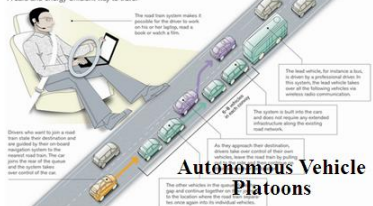
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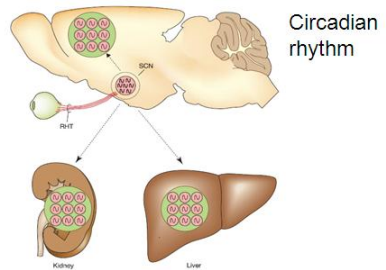
Autonomous Vehicle Platoons

nextbigfuture.com

Other examples



Firefly



Antenna array



Crowd effect

Other examples



Toyota Cartivator
flying car



Drone delivery



gif-finder.com

Formation flying

Multi-Agent System (MAS)

What is an agent?

An intelligent physical or virtual entity that has the following features:

- **autonomy**: self-organized, independent
- **locality**: no global view of the whole system or surrounding environment
- **sociality**: perception of surrounding environment and communication with other agents

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How a multi-agent system work?

- Each agent may have its own goal
- The whole multi-agent system has a global/common objective

⇒ agents need to collaborate with others to achieve the whole MAS objective

Consensus

It is...

equivalent to agreement, e.g., consensus of opinions on an argument. In the mathematical language,

$$\lim_{t \rightarrow t_f} x_i(t) = x^*, \forall i = 1, \dots, n$$

- $x_i(t)$: represents system i
- $t_f > 0$: finite or $+\infty$



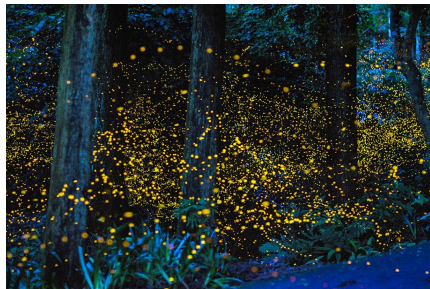
Synchronization

It is...

quite similar to consensus, *but is often used for sync of oscillators' frequencies*. In the mathematical language,

$$\lim_{t \rightarrow t_f} \dot{\theta}_i(t) = \omega^*, \quad \forall i = 1, \dots, n$$

- $\theta_i(t)$: phase of oscillator i
- $\dot{\theta}_i(t)$: frequency of oscillator i
- $t_f > 0$: finite or $+\infty$



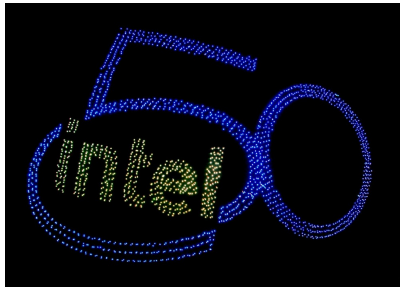
Formation

It is...

used for robotics and vehicles (ground, aerial, underwater). In the mathematical language,

$$\lim_{t \rightarrow t_f} [x_i(t) - x_j(t)] = x_{ij}^*, \forall i, j = 1, \dots, n$$

- $x_i(t)$: represents system i
- $t_f > 0$: finite or $+\infty$



Formation of drones

Flocking, Schooling

They are...

often used for birds, fishes, insects, and animals, and are kinds of *swarm behaviors*. robotics and vehicles (ground, aerial, underwater).



Flock of starlings